Learning the Missing Conditions

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Phase 4 diagnostics for the Metric Stack of Concern

Research-Derived Experiments - controlled L4-parallel harnesses

Abstract

Phase 3 identified several open bottlenecks in the Metric Stack of Concern: language scale, non-enumerative symmetry discovery, learned regime variables, probe value, architecture beyond the mediated-identifiability ceiling, foundation-model-style metric deformation, and topology mediation. This paper reports a controlled Phase 4 diagnostic suite that turns those bottlenecks into seven cheap L4-parallel gates. The allowed claim is bounded: these experiments resolve mechanism choices inside controlled harnesses and identify which claims deserve heavier external validation.

1. Question

The Phase 4 question is whether the missing conditions from the prior arc can be learned rather than handed to the agent: regime partitions, marginal probe value, role-specific mediated structure, semantic metric allocation, and topology/seam factors. The suite is not a foundation-model claim; it is a mechanism-selection filter for the next, more expensive tier.

Field	Value
preset	full
tracks	7 tracks (listed in Table 2)
seeds	48
gpu	L4
rows	1440
claim level	diagnostic controlled-harness result

Table 1. Suite manifest.

2. Gate Results

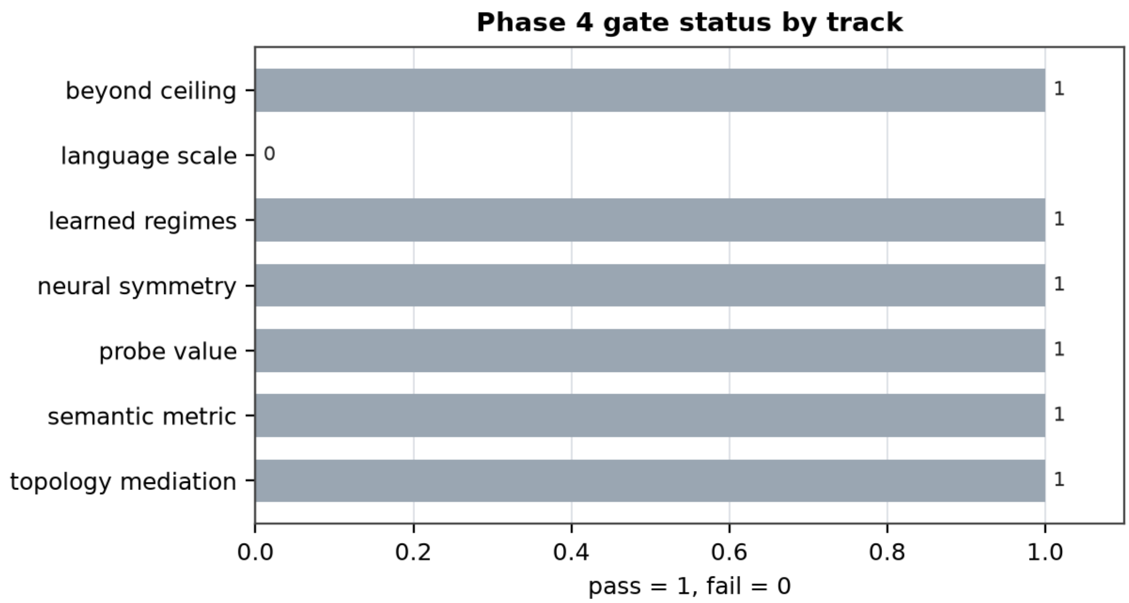

Figure 1. Pass/fail status for the seven Phase 4 gates.

Track	Status	Primary result	Claim level
Language	FAIL	r=0.670; ratio=2.360	mixed: predictive, not 3x causal
Symmetry	PASS	F1=0.945; raw=0.568	closure-constrained generator
Regimes	PASS	boundary=1.000	learned hard partition
Probe value	PASS	VOI red=0.323	marginal value beats error
Ceiling	PASS	role MAE=0.036	role routing beats shared head
Metric	PASS	spec=0.504	controlled semantic deformation
Topology	PASS	seam r=0.921	seam carries mediation

Table 2. Phase 4 gates and allowed claims.

3. Main Findings

Language-scale diagnostics are mixed. Hidden paraphrase geometry predicts log-probability consistency in the large post-coupling condition, but the causal intervention ratio reaches only 2.36x against a 3x gate. The result supports a predictive scale signal while withholding the stronger causal-controller claim.

Non-enumerative symmetry discovery improves when the generator is constrained by group closure. Raw neural proposal is not enough; closure and top-K selection are the load-bearing additions.

The learned-regime gate confirms the Phase 3 diagnosis: smooth approximators can fail at singular boundaries, while a learned hard partition can close most of the oracle gap.

The probe-value track separates current error from marginal information gain. A learned VOI score beats matched-random and current-error allocation, which is the mechanism needed for re-engagement without anxiety.

The role-specific architecture track supports the architectural-ceiling reading: shared mediated heads fail, while disjoint per-role and mixture-style routing recover the mediated coefficients and wrong-history controls fail as they should.

Semantic metric deformation and topology mediation both remain bounded. Value fields move metric density in the controlled semantic harness; topology mediates OOD only through seam consistency, so topology alone is not the causal story.

4. Discovery-Regime Audit

Old regime: Phase 3 could measure concern-like structure in minimal agents but left several variables oracle-specified or architecturally capped. Transition: Phase 4 adds learned regime gates, marginal probe-value targets, role-routed mediated heads, semantic-style metric deformation, and seam-aware topology mediation. Transported evidence: the null-anchor, current-replay, habituation, role-specific ceiling, and metric-deformation claims are preserved as baselines. Rejected alternatives remain visible: raw neural symmetry proposal, current-error probe value, smooth boundary heads, shared mediated heads, and topology without seam consistency.

5. Next Operations

The next tier should spend expensive compute only on tracks that passed these gates: larger real language models for paraphrase action coupling, learned closure-constrained transform generators, learned regime variables in richer homeostatic environments, true VOI probes under regime shifts, role-routed self/world heads, and foundation-model semantic metric deformation. Failed or bounded tracks should be carried forward as controls, not erased.

References

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Brown, J. Weakness, Not Compression. Research-Derived Experiments, 2026.
Brown, J. Future Control Moves Memory. Research-Derived Experiments, 2026.